

Part 2: Batch Processing & Analytics

1. Overview

This part processes the stored ticket data from Part 1 to produce useful analytical reports about events, bookings, seats, and users.

2. Objectives

- Write batch jobs that read data from Part 1 and perform aggregations and basic analysis.
- Demonstrate that the jobs run in parallel across the distributed cluster.
- Combine information from more than one dataset when needed.
- Save the generated results back to distributed storage so they can be used by Part 4.

3. Requirements

- Implement at least 8 batch jobs, including:
 - Total bookings per event.
 - Seat occupancy percentage per event.
 - Total revenue per event.
 - Number of available seats per event.
 - Top 5 most-booked events.
 - Booking statistics by event category.
 - Booking statistics by date.
 - Top 5 users by number of bookings.
- At least 4 jobs must process data from two different datasets (for example, events + bookings or users + bookings).
- At least 2 job must produce a ranked/sorted result, such as Top 5 events or users.
- Jobs must read input from the distributed storage and write output back to it.
- Jobs should be able to run again without breaking or duplicating previous results if re-run.
- The jobs must demonstrate parallel processing across the cluster.
- Provide a simple way to trigger all jobs, such as a script or command.
- Save the output in a clear folder structure so Part 4 can access the results.